constant called universal gas constant.

The above equation can be rearranged to give the ideal gas equation

$$PV = nRT. \quad \text{------ (6.11)}$$

We already know that pressure is expressed in many different units (Table 6.1) hence it is important to know the values of gas constant R in different units as well.

We can calculate R using the equation,

$$R = \frac{PV}{nT}$$

For Conditions in which P is 1 *atm*., volume 22.414 dm^3. for 1 mole at 273.15 K.

$$R = \frac{1\,\text{atm.} \times 22.414\,\text{dm}^3}{1\,\text{mol.} \times 273.15\,\text{K}}$$

$$= \mathbf{0.0821\ dm^3\ atm.\ mol^{-1}\ K^{-1}}$$

Under standard conditions (STP) Where P = 1 bar (10^5 pascal), V= 22.71 × 10^{-3} m^3 for 1 mole of a gas at 273.15 K

$$R = \frac{10^5\ \text{Pa} \times 22.71 \times 10^{-3}\ \text{m}^3}{1\,\text{mol.} \times 273.15\,\text{K}}$$

$$= \mathbf{8.314\ Pa\ m^3\ K^{-1}\ mol^{-1}}$$

$$= \mathbf{8.314 \times 10^{-5}\ bar\ m^3\ K^{-1}\ mol^{-1}}$$

$$= \mathbf{8.314 \times 10^{-2}\ bar\ dm^3\ K^{-1}\ mol^{-1}}$$

$$= \mathbf{8.314 \times 10^{-2}\ bar\ L\ K^{-1}\ mol^{-1}}$$

$$= \mathbf{8.314\ J\ K^{-1}\ mol^{-1}}$$

The ideal gas equation is a relationship between four variables (P, V, T, n). Since it describes the state of any gas, it is referred to as the equation of state of gases.

Let us calculate the pressure exerted by 2 moles of sulphur hexafluoride in a steel vessel of volume 6 dm^3 at 70 °C assuming it is an ideal gas.

We will use the ideal gas equation for this calculation as below:

$$P = \frac{nRT}{V} = \frac{2\ \text{mol} \times 0.0821\ \text{L atm. K}^{-1}.\text{mol}^{-1} \times (70 + 273\ \text{K})}{6\ \text{dm}^3}$$

= 9.39 atm.

Evaluate Yourself

5. A small bubble rises from the bottom of a lake where the temperature and pressure are 8° C and 6.4 *atm.* to the water surface, where the temperature is 25°C and pressure is 1 *atm*. Calculate the final volume in (mL) of the bubble, if its initial volume is 2.1 mL.

6.4 Mixture of gases - Dalton's law of partial pressures

Studies of non-reacting gaseous mixtures showed that in a gaseous mixture each component behaves independently. For a gaseous mixture, it is important to know, how the pressure of individual component contributes to the total pressure of the mixture.

John Dalton stated that "the total

11th Std Chemistry 158-185.indd 166 06-01-2021 15:35:18